Securing Passive Keyless Entry and Start System in Modern Vehicles Based on LF-Band Signal Analysis

Kyungho Joo, Hyo Jin Jo, and Wonsuk Choi

***Abstract*—The low-frequency-band (LF-band) communication in the passive keyless entry and start (PKES) system is basically designed to enable short-range communication (1 to 2 m) through which a key fob determines whether it is in the vicinity of its paired vehicle. However, this short-range communication is vulnerable because it is unable to precisely verify the distance, as the LF-band signals can be easily relayed or amplified. In this article, we present a novel method (named low-frequency fingerprinting, `LOFI`) to detect LF-band signals generated by an attacker. `LOFI` is designed as a subauthentication method that supports existing authentication systems for PKES systems. Through a series of experiments, we demonstrate that `LOFI` effectively detects attacks on the PKES system, achieving an average false positive rate (FPR) of 0.92% and an average false negative rate (FNR) of 0.01% under Non-Line-of-Sight (NLoS) conditions. Moreover, using a physics-based ray-tracing simulation, we analyze detection boundaries against feature impersonation attackers.**

***Index Terms*—Keyless entry system, passive keyless entry and start (PKES) system, relay attack, RF fingerprint.**

I. Introduction

PASSIVE keyless entry and start (PKES) systems allow drivers to conveniently unlock the door without interacting with the key fob. All drivers have to do is touch the car door handle or to push a button on the door to make it unlock. In this system, short-distance communication is the basic technology needed to prevent a situation in which doors unlock when the legitimate key fob is far away from the vehicle. Most modern vehicles are equipped with low-frequency-band (LF-band) communication, whose range is only 1 to 2 m, so that the key fob placed within the communication range ca only transmit response signal to its vehicle. As a result, a vehicle can verify the proximity between the corresponding key fob when it receives a response signal from the key fob. However, in 2011, a signal-relaying attack against the PKES system [1] demonstrated that an attacker can forcibly trigger a key fob that is placed beyond the communication range with the vehicle via relay or amplification of the LF-band signal. The root cause of the signal-relaying attack is that an attacker can inject a malicious LF-band signal as the key fob always accepts any LF-band signal containing a valid binary sequence. In addition to signal-relaying attacks, researchers have also recently demonstrated that weaknesses in a cryptographic algorithm or vulnerabilities in software updates can be used to unlock a vehicle [2], [3], [4]. Similar to a signal-relaying attack, in these attacks, an attacker injects a malicious LF-band signal to a key fob so that it is triggered to transmit valid responses. By analyzing those responses from the key fob, the attacker is able to discover the secret key used for the cryptographic protocol or they can duplicate the key fob. Due to these security flaws, even today, actual cases of vehicle theft caused by weaknesses in the PKES system continue to occur around the world [5], [6], [7], [8].

A class of cryptographic protocols, which is known as distance bounding protocols [9], [10], may be effective preventative measures against relay attacks on the PKES system. Indeed, distance-bounding protocols are implemented in current PKES systems in order for a vehicle to verify proximity to a key fob. Automotive manufacturers usually set a value for the timeout, by which the vehicle is able to verify proximity with the key fob. If the elapsed time is less than the timeout, the vehicle would accept a request for door unlock. However, the narrow-band communication system implemented in the PKES system is not capable of precisely measuring when a signal is received (i.e., time-of-arrival, ToA) to compute the physical distance. This is because approximately tens of nanoseconds are required for a rising edge in the narrow-band communication [11]. During this amount of time, significant errors in measuring ToA would arise because of the channel noise and multipath effects. As a result, there is a large amount of distance error due to the timing error of ToA. Francillon et al. [1], have found out that large values (e.g., tens of ms) were indeed used in modern vehicles to reduce verification error occurrence (while not hindering driver convenience).

To secure the PKES system, which is currently vulnerable to signal-relaying attacks and cryptanalysis, we propose an

Manuscript received 15 July 2024; accepted 22 August 2024. Date of publication 3 September 2024; date of current version 6 December 2024. This work was supported in part by the Institute of Information and Communications Technology Planning and Evaluation (IITP) Grant funded by the Korea Government (MSIT) under Grant RS-2024-00341722; in part by the Development of Cyber Resilience Method for Intelligent Service Robots, 50% and under Grant RS-2020-II200374; and in part by the Development of Security Primitives for Unmanned Vehicles, 50%. *(Corresponding authors: Hyo Jin Jo; Wonsuk Choi.)*

Kyungho Joo is with the School of Software, Soongsil University, Seoul 06978, Republic of Korea (e-mail: khjoo@ssu.ac.kr).

Hyo Jin Jo is with the Graduate School of Information, Yonsei University, Seoul 03722, Republic of Korea (e-mail: hyojin.jo@yonsei.ac.kr).

Wonsuk Choi is with the Graduate School of Information Security, Korea University, Seoul 02841, Republic of Korea (e-mail: wonsuk85.choi@gmail.com).

Digital Object Identifier 10.1109/JIOT.2024.3453920

© 2024 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

RF-fingerprinting method (named LOFI) that verifies LF-band signals from a vehicle based on a characteristics analysis. In our experiment, LF-band signals are measured by a low-cost software-defined radio (SDR). For an antenna, we use the same 3-D loop antenna as the one used in actual key fobs. LOFI achieves an average false negative rate (FNR) of 0.01 % in various Non-Line-of-Sight (NLoS) channel conditions depending on the location of the key fob. According to our results, LOFI produced an average false positive rate (FPR) of 0.92 %. Furthermore, based on a 3-D ray-tracing simulation, we analyze the attack success distance against a strong attacker who can impersonate a legitimate signal. Through this simulation, we show that the attacker only successfully carries out an attack when he/she is located within 1.27 m of the key fob. Our contributions are as follows.

1) We propose an RF-fingerprinting method to secure the PKES system against all types of attacks that have been introduced to date, such as relay attacks, cryptographic attacks, and over-the-air (OTA) attacks.
2) LOFI can fundamentally eliminate or minimize the attack surface of a key fob that acquires and analyzes key fob signals by having the key fob generate its own signals exclusively for legitimate vehicle requests.
3) We analyze which features mainly contribute to detection per attack type, and moreover, based on the correlation between RF impairment factors and the features used in LOFI, we provide concrete reasoning as to why each feature is affected during an attack.
4) We analyze the performance of LOFI assuming the role of various advanced attackers using identical electronic components, high-sample rates, and feature impersonation capabilities. Especially, utilizing a physics-based ray-tracing simulation, we analyzed detection boundaries against feature impersonation attacker. In summary, feature impersonation attacker can only successfully carry out an attack when located within 1.27 m of the key fob.

The remainder of this article is organized as follows. Section II introduces related works and Section III provides background knowledge to understand the design of LOFI. Section IV introduces attacker models and corresponding attack scenarios against the PKES system. Section V presents how LOFI detects the attacks, including the training and verification procedures. Section VI provides extensive evaluation results to show attack detection performance under various attack scenarios and environmental conditions. Sections VIII and IX introduce discussions and conclusion of this article, respectively.

II. Related Work

A. Attacks on the PKES System

Alrabady and Mahmud [12], [13] theoretically analyzed potential attack scenarios that can be occur against PKES systems and proposing possible solutions to prevent the attacks. Francillon et al. [1] pragmatically showed a different attack type on PKES systems, which is the so-called relay attack. The authors performed a comprehensive experiment and proposed two attack scenarios: 1) a wired relay attack and 2) a wireless relay attack. In addition, Li [14] showed a digital relay attack in which attackers can demodulate/decode the binary sequence. Compared to previous attack scenarios [1], the attackers in Zeng et al. exchange the binary sequence rather than the analog signal. Wouters et al. [3], [4] introduced attacks on cryptographic algorithms and software update mechanisms to unlock vehicles. Researchers revealed that the PKES systems used by major commercial car manufacturers, including Toyota, Hyundai, and Tesla, use vulnerable cryptographic algorithms like DST40 and DST80. Wouters et al. [2] also demonstrated that an attacker can inject malicious firmware to a key fob, exploiting a vulnerability of an OTA update. After compromising the key fob, the attacker requests a valid unlock command from the key fob and unlocks the vehicle.

B. RF Fingerprinting

RF fingerprint systems aim to identify (or verify) RF devices based on signal characteristics during communication processes at the physical layer. Depending on the signal region used to extract fingerprints, the RF fingerprint system can be categorized as a transient-based or modulation-based approach. Transient-based approaches use the turn-on/off transient of an RF signal for device identification whereas modulation-based approaches extract features from a detected symbol, such as I-Q imbalance, phase offset, or carrier frequency offset (CFO). The target devices of the RF fingerprint system include VHF transmitters [15], IEEE 802.15.4 transceivers [16], Bluetooth transceivers [17], [18], and HF/UHF RFID transponders [19], [20]. A large portion of these methods employ a high-end vector signal analyzer and extract features using principal components analysis (PCA), which requires a significant amount of computational cost. An RF fingerprint system that can detect a fake base station (FBS) has been also introduced [21]. In addition, several deep-neural-network-based approaches have emerged in recent years. Youssef et al. [22] proposed a multistage training procedure to investigate the classification and identification problem of the proprietary OFDM protocol similar to IEEE 802.11a/g. He and Wang [23] proposed specific emitter identification (SEI) schemes based on long short-term memory (LSTM). Shen et al. [24] employed the deep learning model as a feature extractor to identify LoRa devices. Finally, Qian et al. [25] employed the convolutional neural network (CNN) and a PCA method to identify automatic identification system (AIS) signals from actual ships. We defer to [26] and [27] for comprehensive explanation of RF fingerprinting methods.

III. Background

In this section, we provide background knowledge to aid with understanding of the following sections. We first describe the system architecture of the PKES system. Next, we introduce the concept of device-specific RF impairments that can be used as unique characteristics of individual RF devices.

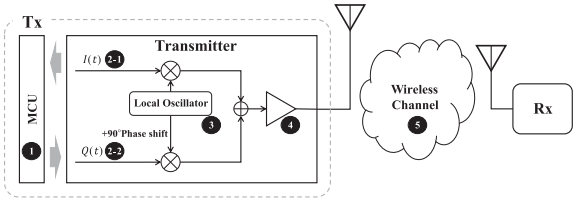


Fig. 1. Impairments occurring in wireless digital communication system.

A. PKES System

The PKES system was developed for driver convenience. When a key fob is in the vicinity of a vehicle, communication occurs without any interaction of a driver, and LF-band communication is used to verify the proximity between a vehicle and a key fob. The PKES system is designed to limit the range of LF-band communication to one to 2 m, while UHF-RFID, Bluetooth low energy (BLE) in an industrial, scientific, and medical (ISM) band is used between a key fob to a vehicle with a more limited approximate range of 100 m. Accordingly, the vehicle periodically transmits a scanning signal to verify the proximity. The key fob then transmits a response if the key fob receives the scanning signal. After proximity verification, the vehicle should authenticate the key fob. When a driver touches a door handle, the vehicle transmits a challenge signal. If the key fob has a preshared secret key with the vehicle, it finally transmits the correct response to the challenge signal, and as a result, the vehicle doors are unlocked. To verify a proximity of a key fob, automotive manufacturers usually set a value for timeout. The vehicle would accept a request for door unlock only when it receives a valid response within the timeout. However, Francillon et al. [1] have found out that tens of ms values are indeed used to reduce an error rate in modern vehicles.

B. Device-Specific RF Impairments

In modern digital communication, a binary sequence is modulated as a form of a continuous signal, and the modulated signal passes through different device-specific RF impairments, such as clock difference, I-Q imbalance, LO offset, and power amplifier. In general, RF impairments are compensated on the receiver side for correct demodulation, but RF-fingerprinting approaches are motivated by the fact that these device-specific properties can be used as a device fingerprint. Fig. 1 illustrates an example of a block diagram for a quadrature amplitude modulation (QAM) transmitter, in which it can be seen where each type of RF impairment (indexed by circled numbers) is generated.

❶*System Clock (SC):* The microcontroller unit (MCU) processes binary information and controls the transmitter to generate an analog signal. The instruction that determines the operation of the MCU is performed based on its own clock signal, which has a unique offset for each device due to hardware imperfections in the inductor and capacitor that constitute the oscillator circuit [28].

❷ *I-Q Imbalance:* In digital communication, each symbol is mapped into a combined form for an in-phase (I), quadrature (Q) component. Fig. 1 shows the block diagram of a generalized quadrature modulation. An ideal I-Q modulator and demodulator provide two orthogonal channels for real and imaginary parts of a complex signal, which are known as the in-phase and quadrature components, respectively. However, I-Q imbalances occur due to mismatches between parallel sections, finite tolerances of the capacitor, and resistor values used to implement analog components.

❸ *LO Offset:* A baseband signal is mixed with a sinusoidal carrier frequency (i.e., operating frequency) that is generated by the local oscillator (LO). The LO has a unique frequency offset with respect to the ideal carrier frequency due to the hardware imperfections in the resistor and capacitor. In addition, the CFO is caused by the Doppler effect as the transmitter or receiver moves.

❹*Nonlinearity:* A power amplifier is an essential component in modern communications systems that provides a sufficient transmit power to overcome any propagation loss between the transmitter and receiver [29]. However, the power amplifier may distort the input signal, induce additional noise within the signal frequency range, or generate unwanted harmonic frequencies in adjacent channels. If the input signal level is located at the nonlinear region of the power amplifier, nonideal characteristics appear in the form of harmonic frequencies, intermodulation distortion, phase distortion, or adjacent channel interference [30].

❺*Channel Impairment:* As a promising property for attack detection, we consider channel impairment along with device-specific impairment. We note that the channel impairment does not represent the characteristics of an RF device. However, it can be employed to detect an adversarial channel created by an attacker depending on the attack scenario. When the RF signal propagates through space, the frequency band of the RF signal determines the amount of attenuation, refraction, diffraction, and reflection. When attackers relay an attack signal through a different frequency band, the attack signal experiences a different channel effect than a legitimate signal.

IV. Attack Model

In this section, we outline hypothetical attackers on the PKES system with specific regard to how each attacker may inject an attack signal so that a key fob responds and unlocks vehicle doors. In addition to this, we present three attack scenarios for the easy understanding of `LOFI`.

A. Attacker Models

In the attack models, the main objective of the attackers is to unlock vehicle doors by exploiting vulnerabilities of the PKES system. For simplicity, we exclude all other functions of a key fob that an attacker could exploit, such as an engine ignition. Since attack signals are generated in different ways, we consider two types of attackers in our model: 1) an *analog* attacker and 2) a *digital* attacker.

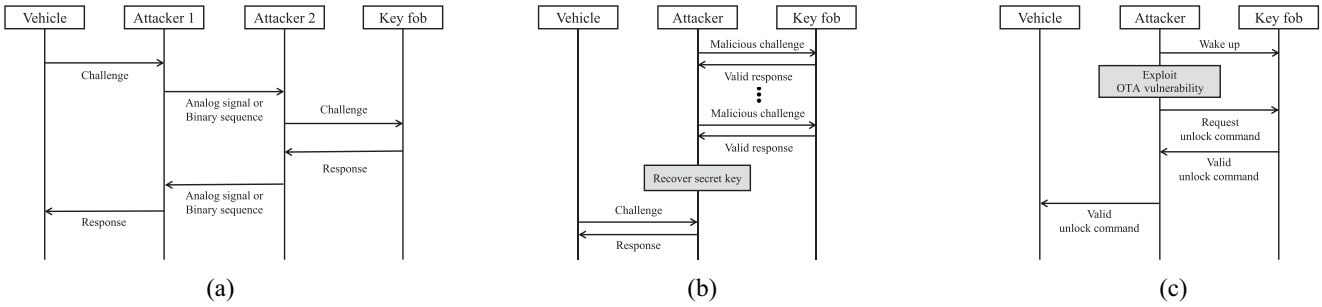


Fig. 2. Attack scenarios: (a) Relay attack [1], [31], (b) Cryptographic attack [3], [4], and (c) OTA attack [2].

Analog Attacker: The analog attacker only uses analog components, such as a frequency mixer, an amplifier, or an analog filter, to attack the PKES system. Since the analog attacker does not have to understand digital signal processing, the attack is simpler and easier than others. The analog attacker up/down converts a legitimate signal generated from a key fob to a specific frequency band using a mixer. This frequency conversion may increase RF antenna efficiency so that the communication distance can be extended. For reliable signal injection, analog attacker amplifies the LF-band signal strength using an RF amplifier and filter out unwanted noise using an analog filter.

Digital Attacker: As an advanced attack, the digital attacker is assumed to be able to perform any method of digital signal processing on an LF-band signal generated from a vehicle. When the attacker completes digital signal processing, a binary sequence is generated. Accordingly, the digital attacker modulates the LF-band signal using binary sequence as the final step of attack signal injection. To modulate the LF-band signal, the attacker must be equipped with an RF device that is able to generate an LF-band signal. The digital attacker can also have analog components similar to an analog attacker for reliable signal injection. In the following section, we introduce three different attack scenarios categorized by attacker type.

B. Attack Scenarios

Based on the attacker models, we provide three attack scenarios: 1) a *relay attack*; 2) a *cryptographic attack*; and 3) an *OTA attack*, as shown in Fig. 2. In the relay attack scenario, both analog attackers and digital attackers can negatively alter the PKES system, however, in the other scenarios, only the digital attacker is capable of attacking the system.

1) Relay Attack: Two attackers are required to carry out a relay attack. One attacker must be located near the target vehicle, while the other is located near the key fob. The attacker at the vehicle forwards the LF-band signal to the other attacker through either a wired or wireless transmission method. Although a wired signal-relaying attack can be implemented with inexpensive RF devices, antennas, cables, or power amplifiers [1], attackers may conduct a wireless signal-relaying attack in practice to avoid obstacles or surveillance system. With this in mind, we only focus on the wireless signal-relaying attack in our relay attack scenario as it fits into actual theft cases [5], [6], [7], [8]. According to the attacker model, we introduce two attacks, an analog relay attack and a digital relay attack as follows.

Analog Relay Attack: The analog attacker at the vehicle converts the LF-band signal to a high-frequency band by mixing a sinusoidal signal $c(t)$. The attacker employs a frequency synthesizer that generates a stable sinusoidal signal as the input to a frequency mixer. The frequency mixer outputs the up-converted LF-band signal that can be represented as the mixed form of an LF-band signal $m(t)$ and sinusoidal signal $c(t)$ (i.e., $m(t) \otimes c(t)$). Since the size of the transmit/receive antenna depends on the wavelength of the signal (inversely proportional to operating frequency), the up-conversion process allows the attackers to use a small antenna to forward the attack signal more effectively. For reliable transmission, the vehicle-side attacker can also employ a power amplifier. A key fob-side attacker mixes the received signal with the sinusoidal signal of the same carrier frequency ($c(t) = \cos(2\pi f_c t)$) used in the up-conversion to remove $c(t)$ and down-convert to the LF-band. Identical with the up-conversion process, the down-conversion process can be accomplished by a signal mixer and frequency synthesizer. Now, the attacker can extract the LF-band signal $m(t)$ and inject it into the key fob. Since the key fob considers the received LF-band signal as legitimate, it transmits the corresponding response signal through the UHF band. As the transmission range of the UHF band is at most 100 m, the vehicle directly receives the UHF signal and is unlocked.

Digital Relay Attack: A digital attacker at the vehicle extracts the binary sequence from the LF-band signal and forwards it to the attacker near the key fob using a variety of digital communication techniques (e.g., Wi-Fi, LoRa, and Zig-bee). Since a digital communication process filters out various noise factors caused by the RF components and the wireless channel, attackers can accomplish an attack at much longer distances compared to analog relay attacks. After extracting the forwarded binary sequence, the attacker near the key fob modulates and injects the LF-band signal into the key fob. Although digital relay attackers can easily carry out an attack without alerting their victim (i.e., the owner of the target vehicle), attackers must minimize the digital communication processing delay to the predefined time out. Recently, researchers have shown that attackers are able to meet the time out with inexpensive hardware components [31]. When the key fob responds to the attack signal, the attacker near the key fob also forwards the binary sequence extracted

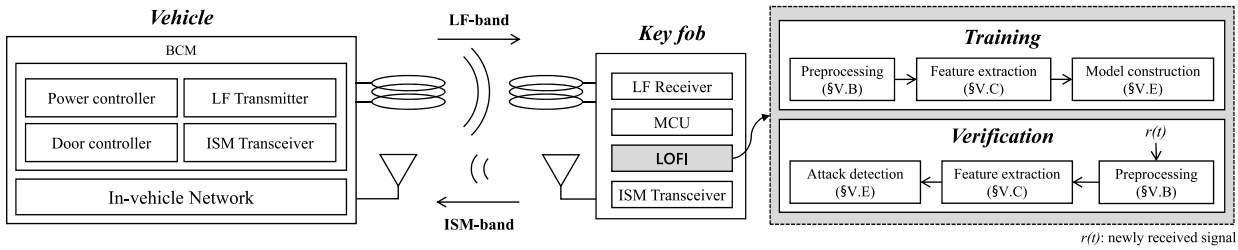


Fig. 3. Design overview.

from the UHF signal to the attacker near the vehicle. Finally, the attacker near the vehicle modulates and injects the UHF signal into the vehicle. When the vehicle receives the UHF signal within the predefined time out, the vehicle verifies that the UHF signal was transmitted from the legitimate key fob.

2) Cryptographic Attack: In some instances, a vulnerable cryptographic algorithm could be used by the PKES system. To compromise a preshared secret key used in the vulnerable cryptographic algorithm, an attacker should have a set of valid challenge-response signals generated from a vehicle and its key fob. For this purpose, the attacker may inject a malicious LF-band signal (i.e., a challenge) into the key fob so that they can obtain a valid UHF band signal (i.e., a response) from the key fob. Indeed, it is widely known that the PKES systems of Toyota, Hyundai, Tesla and other major commercial car manufacturers use vulnerable cryptographic algorithms like DST40 and DST80 [3], [4].

3) OTA Attack: In addition to a cryptographic attack, some vulnerable software updates for the key fob's firmware could be exploited by an attacker. If the firmware of the key fob is manipulated with malicious software through an update, the attacker could compromise the key fob and unlock the vehicle. In an OTA attack, an attacker injects an LF-band signal to *wake the key fob up* by impersonating a valid body control module (BCM) of the target vehicle. Then, the attacker initiates the OTA update session and pushes the malicious firmware to the target key fob through an update channel, such as BLE communication. Once a key fob has been updated and compromised, the attacker forces it to transmit a door unlock signal, which the attacker can then capture. As a result, the attacker can use the captured signal to unlock the vehicle. Indeed, to this end, Wouters et al. [2] recently demonstrated how an attacker can update the firmware of a key fob with a vulnerable OTA update. The authors showed that the Tesla Model X could be unlocked successfully via a door unlock signal from the compromised key fob.

V. LOFI

We describe the design overview of our method, `LOFI`, including the system architecture and overall workflow. Next, we elaborate on how `LOFI` detects an attack on the PKES system.

A. Design Overview

The key fob is an integrated circuit, including an LF-band receiver, an MCU, and the ISM band transmitter. `LOFI` is designed to be simply implemented in a key fob without any additional hardware components. When `LOFI` detects an attack, it alerts the key fob, which, accordingly, does not transmit a valid response for door unlock. Fig. 3 illustrates the overall system model and workflow of `LOFI`.

`LOFI` verifies an LF-band signal that is generated from the vehicle and received by the key fob. Since `LOFI` uses a classification algorithm for verification, `LOFI` has two phases: 1) a *Training* phase and 2) an *Verification (i.e., Testing)* phase. In the training phase, a classifier is constructed, and the parameters for normalization are computed—all of which are used in the subsequent *Verification* phase. The normalization parameters represent distribution characteristics of the training data and adjust the differences between the output distribution of individual devices. In the *Verification* phase, `LOFI` detects attacks defined in our attack model in Section IV. Preprocessing and feature extraction are conducted on a newly received LF-band signal, and a set of the extracted features is used as input data for the trained classifier. Based on the classifier output, `LOFI` determines whether the received LF-band signal was transmitted from a legitimate vehicle. When a case is determined to be invalid, the challenge message is rejected, and the key fob does not transmit a valid response for door unlock.

B. Preprocessing

In the preprocessing procedure, we amplify an LF-band signal using a low-noise amplifier (LNA) and remove high-frequency noise by using a low-pass filter. Since on-off-keying (OOK) modulation is used in LF-band communication, we demodulate the filtered signal by computing the magnitude (i.e., envelope detection). $s[t]$ is denoted by the LF-band signal from which features are extracted. For further feature extraction, we compute the magnitude of $s[t]$ and denote as $d[t]$. Since the OOK modulation represents a binary sequence as the presence or absence of a carrier wave, the magnitude of $s[t]$ (i.e., $d[t]$) is a square wave that represents a binary sequence. As a last step of the preprocessing chain, we divide $s[t]$ and $d[t]$ by their respective root-mean-square (RMS) to minimize feature variation due to the received signal strength (RSS), which denoted as $s_{\text{rms}}[t]$ and $d_{\text{rms}}[t]$, respectively.

C. Feature Extraction

We extract features from the preamble field of a received LF-band signal. The preamble field contains predefined static value which enables clock synchronization between transmitter

and receiver. Since the preamble field always shows the same pattern, we are able to extract meaningful features that represent the uniqueness of a transmitter. In wireless digital communication, the preamble field is designed for accurate time synchronization between a transmitter and receiver. Accordingly, the preamble field always exists in LF-band communication. We extract two types of features that are categorized by *impairment features* and *statistical features*. Compared to the recent studies that employing deep neural network (DNN) architecture for automated feature extraction [32], [33], [34], [35], [36], [37], `LOFI` is designed to manually extract all possible features and select appropriate features through a feature selection procedure. This is because key fob devices are usually resource-constrained embedded boards that are incapable of implementing DNNs.

1) Impairment Features: We extract impairment features that are known to represent distinct attributes of an RF device. It should be noted that the *impairment features* are considered errors to be compensated at the receiver because it causes performance degradation in wireless communication. However, we embrace the impairment features used to verify whether the received signal is transmitted from the legitimate vehicle. We extract the following impairment features from $s[t]$ and $d[t]$.

CFO: After the preprocessing procedure, we extract a CFO feature from $s[t]$. As each RF transmitter has a nonideal (i.e., deviated from 134.2 kHz) carrier frequency due to imperfections in its LO, the RF receiver should operate on a wide band to reliably receive signals from the transmitter in a real communication system [38]. We define the CFO as the frequency at which the highest amplitude value exists as follows:

$$\text{CFO} = \arg\max_{f}(S[f]) \tag{1}$$

where $S[f]$ represents the fast Fourier transformation (FFT) of $s[t]$.

SC: We extract a SC from $d[t]$, which represents device-inherent characteristics. As described in Section III, we know that each MCU operates on a slightly different clock deviated from an ideal value. As a result, a preamble field $d[t]$ is affected by a SC in the time and frequency domains. Specifically, the pulse width of $d[t]$ represents the hardware clock characteristic of an MCU. Since the pulse width and clock signal have an inverse relationship, a device with a faster clock has a wider pulse width. As a square wave contain a wide range of harmonics, it can be seen that several dominant peaks (odd harmonics) exist in the frequency domain [39], [40]. We compute the SC as follows:

$$\text{SC} = \frac{1}{\max_f |D[f]|} \tag{2}$$

where $D[f]$ represents the fast Fourier transform result of $d[t]$. We considered additional RF impairment features for `LOFI`, such as DC *offset* and SNR. However, those features proved ineffective; accordingly, we ultimately exclude those features from our evaluation.

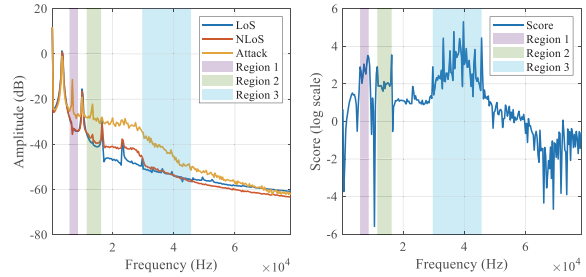


Fig. 4. Spectrum analysis for spectral feature extraction, (left): energy distribution of a demodulated signal $d[t]$ in the frequency domain, and (right): candidate score as a function of a frequency component.

2) Statistical Features: For better detection, we employ statistical features that support the *impairment feature* set. The statistical features represent the statistical characteristics of $d[t]$ and $s[t]$ in the time and frequency domains. We extract statistical features employed in previous studies [18], [21], [39], [41], [42], [43], [44] that identifying specific devices by analyzing physical-layer signals. However, rather than extracting statistical features from the whole frequency range, we determine a specific frequency range rather than a whole range to minimize the number of samples to be computed. With a set of collected signals $d[t]$ under LoS conditions, NLoS conditions, and attack scenarios, we perform FFT on each $d[t]$ and compute an average amplitude for each frequency. We averaged the FFT output from whole vehicle models that are employed in our evaluations (Sections VI and VII). Therefore, all vehicle models employed identical frequency components for feature extraction. Fig. 4 shows the energy distribution of $d[t]$ in the frequency domain. We search for a frequency region where the difference of amplitudes between the LoS and NLoS conditions is small but the LoS and attack signal is large. We use the following equation for the metrics to measure the difference of amplitudes:

$$\text{score}_i = \frac{|D^i_{\text{los}}[f] - D^i_{\text{attack}}[f]|}{|D^i_{\text{los}}[f] - D^i_{\text{nlos}}[f]|}. \tag{3}$$

D^i_{loS} and D^i_{nloS} refers to the ith frequency component of $D[f]$ under LoS and NLoS channel condition without attack, respectively. Likewise, D^i_{attack} refers to the ith component of fast Fourier transform of $d[t]$ under attack scenario.

Based on the score_i, three different frequency ranges are found where the result of the metrics is greater than 100. Each frequency range is highlighted in Fig. 4. In Region 1 and Region 2, we observed unintentional harmonic components due to the nonlinearity of the power amplifier employed in the attack device. Although there are no harmonic components in Region 3, the attack signal has a higher energy level than a legitimate signal. Above Region 3, the energy difference between a normal signal and an attack signal can be seen as negligible due to the digital filter effect. We excluded the frequency component that is more than 70 kHz since its score is less than zero. Based on these observations, we extract a set of statistical features derived from the three frequency regions. As a result, we can reduce the 97.56% frequency component (78/3199) at the feature extraction phase, which impacts execution time and energy consumption (Section VII-C).

TABLE I
SELECTED PARAMETERS FOR TUNING IMPAIRMENT FACTORS

Impairments	Unit	Range
I-Q imbalance	dB	±10
Path loss	dB	10 to 20
System clock offset	µs	±200
Local oscillator offset	kHz	±10

TABLE II
CORRELATION BETWEEN IMPAIRMENT FACTORS AND AFFECTED FEATURES: *Spec.* MEANS A FEATURE IS EXTRACTED FROM THE FREQUENCY DOMAIN. SUBSCRIPT *s* AND *d* MEANS A FEATURE IS EXTRACTED FROM $s[t]$ AND $d[t]$, RESPECTIVELY

Impairment	Number	Features
I-Q imbalance Path loss System clock	10	Spec.Brightness$_d$, Spec.Spread$_d$, Spec.Flux$_d$, **Spec.Centroid$_d$**, StandardDeviation$_d$, Spec.Kurtosis$_d$, Spec.Flux$_d$, Spec.Irregularity$_d$, **Spec.Entropy$_d$**, Spec.Skewness$_d$
Path loss I-Q imbalance	13	DC offset$_d$, Max$_d$, **Spec.Flatness$_d$**, IQR$_s$, StandardDeviation$_s$, Max$_s$, Mean$_d$, Spec.Brightness$_s$, RMS$_s$, Min$_s$, RMS$_d$, IQR$_d$, Kurtosis$_d$
Path loss System clock	2	Spec.Rolloff$_d$, SNR$_d$
Path loss Local oscillator offset	2	Spec.Centroid$_s$, Mean$_s$
I-Q imbalance	2	Kurtosis$_s$, Mean$_s$
Path loss	9	Skewness$_d$, Spec.Flatness$_s$, Spec.Rolloff$_s$, Spec.Entropy$_s$, Min$_d$, Spec.Kurtosis$_s$, SNR$_d$, Spec.Skewness$_s$, Spec.Flux$_s$
System clock	3	**SC**, NNC$_s$, **Power$_d$**
Local oscillator offset	1	**CFO**

D. Impact of RF Impairments on the Features

An RF impairment can have an impact on multiple features, including impairment features and statistical features. To clarify the relationship between the RF impairment and extracted features, we analyze the correlation between them based on the *p-value* [45] while varying the degree of an impairment (e.g., noise and offset level). If a *p-value* of a correlation coefficient is less than 10^{-3}, we determine the corresponding impairment and feature to be correlated. We generate a preamble field consists of ten ASK-modulated symbols (i.e., [1 0 1 0 1 0 1 0 1 0]) and simulated the RF impairments according to the parameters represented in Table I. At this point, we note that other parameter range is also possible because we determine whether the RF impairment and extracted feature is *correlated or not*. Table II shows the correlated pairs of individual features and impairments. The first column in Table II shows the RF impairments correlated with the features in the third column. Specifically, for the first row of the table, the RF impairment (i.e., I-Q imbalance, Path loss, and SC) impacts the ten features in the third column. This implies that those features can be an effective feature when the attack signal has different characteristics due to I-Q imbalance, Path loss, or SC. In Section VI-E, combined with the result of feature importance in corresponding attacker model, we provide concrete reasoning on which RF impairment manipulate the signal characteristics during the attack. For example, when *CFO* is the most effective feature to detect a certain attack, we can analyze that the characteristic of LO in attacker device allows LOFI to detect the attack.

Algorithm 1 Verification

function VERIFICATION ($s[t]$: RECEIVED SIGNAL)
  suspiciousCnt $= 0$ ▷ Initialize suspicious count
  $d_{\text{rms}}[t], s_{\text{rms}}[t] \leftarrow$ preprocessing $(s[t])$
  $F \leftarrow$ featureExtraction $(d_{\text{rms}}[t], s_{\text{rms}}[t])$ ▷ F: feature set
  $O \leftarrow \mathbb{C}\ (F)$ ▷ O: output score, $\mathbb{C}$: Anomaly detector
  $O \leftarrow \frac{|O-\mu|}{\sigma}$ ▷ Normalize output score
  while i $<$ N **do** ▷ N: Number of received signals
    if $O_i > \Gamma$ **then** ▷ Γ: threshold
      suspiciousCnt$+ = 1$ ▷ Increase suspicious count
    end if
  end while
  if suspiciousCnt $> \left\lfloor \frac{N}{2} \right\rfloor$ **then**
    Reject ▷ Attack
  else
    Accept ▷ Normal
  end if
end function

E. Training and Verification

Training: We construct a classifier that is trained with legitimate signals. In practice, the number of attack signals, including all potential attack scenarios, to be used for the classifier construction, may be insufficient. A central idea in LOFI is to train a model of normality from legitimate signals so that any anomalies can be detected by a deviation from our model [46], [47], [48], [49], [50], [51], [52], [53], [54]. We employ well-known, one-class (OC) classification algorithms, as in the OC support vector machine (OC-SVM), Gaussian mixture model (GMM), and k-nearest neighbor (k-NN) algorithms. The OC-SVM objective is to find a hyperplane that separates the data from the legitimate data in the feature space with the maximum possible margins. The k-NN is a distance-based algorithm without building a model but evaluates new test data with respect to legitimate data. The GMM fits a multivariate Gaussian distribution to the training data set and evaluates the log-likelihood of test data according to the trained model.

Verification: With a trained OC classifier, LOFI verifies a newly received LF-band signal by checking whether the signal is classified by the valid class or not. In the PKES system, multiple packets from a vehicle are sent because multiple rounds are observed in the communication protocol. To minimize a false-alarm rate, we determine an attack when more than a half number of LF-band signals are classified as invalid. LOFI is designed to detect an attack if the number of invalid classifications exceeds $\lfloor N/2 \rfloor$, where N is the number of total observed packets. Algorithm 1 illustrates the training and verification of LOFI.

VI. EVALUATION

LOFI is evaluated in five actual vehicles, considering real-world channel conditions and environmental variations.

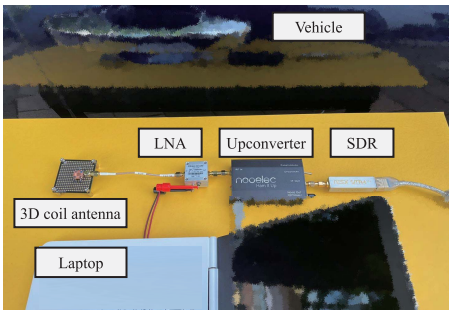


Fig. 5. Experimental setup for signal acquisition under the LoS channel.

TABLE III
VEHICLE MODEL SPECIFICATIONS

Vehicle model	Type	Year
Vehicle A-1		2020
Vehicle A-2	Compact sedan	2020
Vehicle A-3		2020
Vehicle B	Compact SUV	2014
Vehicle C	Luxury sedan	2018

A. Experimental Setup

Actual Vehicle: We perform a series of experiments on five vehicles of three models, including three compact sedans (Vehicle A), one compact sports utility vehicle (Vehicle B), and one luxury sedan (Vehicle C). We specially employ the three vehicles for Vehicle A to show that distinct features can be observed from three vehicles of the same model manufactured in the same year (2020). All models are equipped with the PKES system using LF/UHF band communication. Table III shows information for the vehicles used in our evaluation. In our setup, we observed all vehicle models transmit 4 LF-band signals during the challenge-response phase (i.e., $N = 4$). Therefore, LOFI raises an alarm when it detects more than 3 attack signals.

Signal Acquisition and Injection: An antenna should have a proper hardware architecture according to operating frequency, which maximizes antenna gain [55]. In general, a key fob has two antennas, one for ISM band communication and one for LF-band communication. A PCB-printed antenna and a 3-D coil antenna are used for ISM band and LF-band communication, respectively. To acquire LF-band signals, we assemble a 3-D coil antenna (ZSUCORE), which is used for the PKES system of the real vehicles [56] on a perfboard as shown in Fig. 5. To inject attack signals, we used a loop antenna (Tecsun AN-200). Next, we employ Nooelec NESDR which is a low-cost SDR for signal acquisition. Nooelec NESDR provides an analog-to-digital converter (ADC) with a resolution of 8 bits. Since Nooelec NESDR only measures an RF signal in the 500 kHz to 1.75 GHz range, we need to use an upconverter; accordingly, the *Ham it up* upconverter [57] brings an LF-band signal into the tunable range for Nooelec NESDR. In addition, we sample RF signals at 2 MS/s. Fig. 5 shows the experimental setup for signal acquisition.

ML Algorithms: We employ MATLAB codes for the OC-SVM, k-NN, and GMM algorithms. These algorithms are used for attack detection in LOFI. In our evaluation, a radial basis function (RBF) is used as a kernel function of OC-SVM. For the k-NN algorithm, the standardized Euclidean distance is applied to compute a distance of features. For the GMM algorithm, we set the number of Gaussian distributions as one and initialize parameters for the expectation-maximization (EM) algorithm employing the k-means++ algorithm, a default setting in MATLAB implementation. For each algorithm, we collect 100 LF-band signals transmitted from a vehicle at a distance of 50 cm under an LoS channel condition.

TABLE IV
ASSIGNED THRESHOLD FOR ALGORITHM AND VEHICLE MODEL

Vehicle Model	Algorithm		
	OC-SVM	k-NN	GMM
Vehicle A	7	100	50
Vehicle B	1.5	81	10
Vehicle C	9	90	100

Performance Metrics: To show the overall performance of LOFI, we calculate a FPR and a FNR. FPR represents the scenario in which a legitimate signal is considered an attack, whereas FNR represents the case in which an attack signal is considered legitimate.

Threshold Parameter: Since LOFI is designed with a OC classification algorithm, a threshold is required to determine whether the received signal is legitimate. Table IV shows the thresholds used for each algorithm and vehicle model. It can be seen that the same threshold is shared within the same vehicle model, which is an important property in practice.

B. Feature Selection

Without a feature selection, performance loss can occur due to the redundant feature set. Moreover, when it comes to execution time, redundant features cause additional processing delays for feature extraction. Because the feature selection problem is NP-hard, we consider a suboptimal search strategy. We employ minimum redundancy maximum relevance (MRMR), multicluster feature selection (MCFS), Fisher, and Laplacian algorithms provided by a MATALB package [58]. We select the six common features from the results as follows: 1) *SC*; 2) *CFO*; 3) Power_d; 4) Spec.Centroid_d; 5) Spec.Entropy_d; and 6) Spec.Flatness_d. We note that the selected features are commonly used regardless of ML algorithms and vehicle models.

C. Detecting Analog Attacker

Two analog attackers use analog RF components to convert an LF-band signal to an ISM band signal, and conversely. In the conversion process, a signal is mixed with a ISM band carrier signal generated from a LO. We use a frequency mixers, ZX05-U432H-S+ [59] and ZX05-43-S+ [60], for up and down conversions, respectively. To generate a carrier signal, we use a frequency synthesizer, ADF4355-3 [61], which includes a voltage controlled oscillator (VCO), a phase locked loop (PLL), and a temperature controlled crystal oscillator (TCXO). In addition, for attack signal injection, a power amplifier is employed to forward and inject an attack signal. We set a distance between two attackers as 10 m under LoS

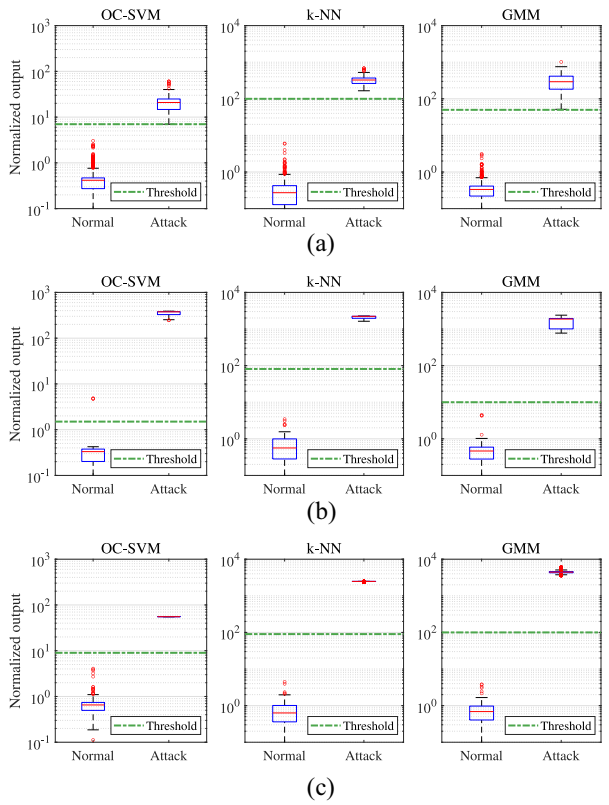


Fig. 6. Output distribution of each classifier during an analog relay attack. (a) Output distribution of Vehicle A-1. (b) Output distribution of Vehicle B. (c) Output distribution of Vehicle C.

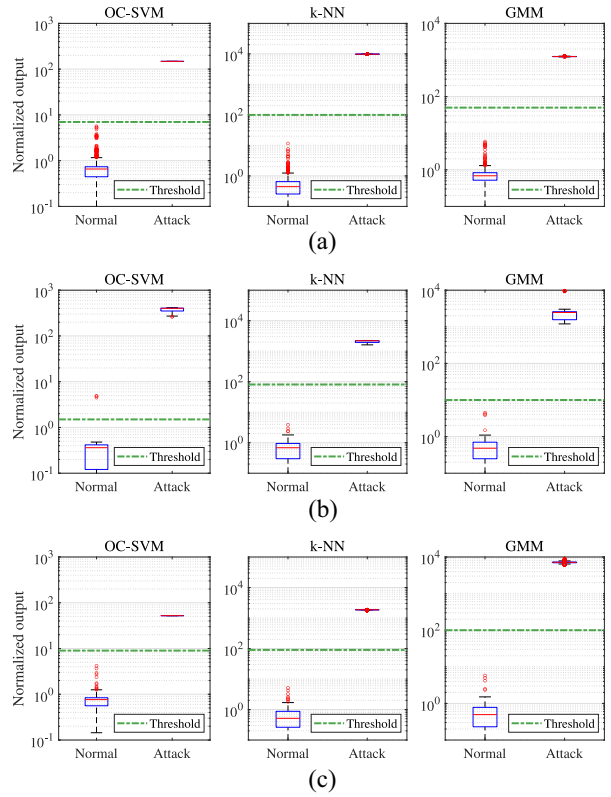


Fig. 7. Output distribution of each classifier under a digital relay attack. (a) Output distribution of Vehicle A-1. (b) Output distribution of Vehicle B. (c) Output distribution of Vehicle C.

condition. To provide the best channel conditions for an attacker, we position the attacker's antenna and our 3-D coil antenna in close proximity to each other (within 10 cm). Next, we sampled 200 attack signals and used these as input to the classifier. Fig. 6 shows the output distribution of ML algorithms in Vehicle A-1, B, and C. At the static thresholds, an average FPR of 0 % and an average FNR of 0 % are seen from Vehicle A-1, Vehicle B, and Vehicle C.

D. Detecting Digital Attacker

We presume that a digital attacker conducts one of the three outlined attack scenarios (i.e., a digital relay attack, a cryptographic attack, or a OTA attack scenario), as mentioned in Section IV. A digital attacker basically uses an RF device to generate and inject an attack signal of a valid binary sequence. Accordingly, we consider two types of devices through which a digital attacker generates and injects an attack signal.

1) SDR Device: Given a binary sequence, a digital attacker should modulate an RF signal. We employ USRP X310 with the LFTX daughterboard, which enables the USRP to modulate LF-band signals [62], along with a loop antenna and a RF power amplifier. We placed the 3-D coil antenna and the loop antenna of an attacker close to each other to minimize path loss. Like detecting an analog attacker, a classifier is trained with 100 legitimate signals, and then 500 attack signals are tested. The attack signals are generated by the USRP at a sample rate of 2MS/s. Moreover, as introduced in [63], we generate attack signals with the same parameters as the target PKES system, such as carrier frequency, bit time, and binary sequence. Fig. 7 shows the output distribution of OC-SVM, *k*-NN, and GMM according to vehicle model. Compared to an analog attack, the signal characteristics of digital attack signals more clearly deviate from a legitimate signal. As a result, `LOFI` has an average FPR of 0 % and an average FNR of 0 % across the three vehicle models.

2) Same Vehicle Model: Wouters et al. [2] demonstrated that an attacker can reprogram a BCM module so that an OTA attack is performed on a target key fob. As an advanced digital attacker, we further assume an attacker who uses the same RF device as the target vehicle model. To mount this advanced attacker, we use vehicles that are the same Vehicle As the target vehicle. That is, LF-band signals from Vehicle A-2 and A-3 are used as the attack signals. Fig. 8 shows the output distribution of Vehicle A-1, while assuming the transmitters of Vehicle A-2 and Vehicle A-3 as the attack device. Compared to the output distribution of USRP, as shown in Fig. 7(a), the output distributions of Vehicle A-2 and Vehicle A-3 are more close to that of USRP in Fig. 8. From this result, we observe that an attacker signal generated by vehicles of the same model has more similar features compared to an attack signal generated by the USRP. However, `LOFI` is able to correctly differentiate an attack signal even if it is generated from the same vehicle model. `LOFI` has an average FPR of 0 % and average FNR of 0 % in regards to detecting a digital attacker using the same RF device.

3) SDR Device With High-Sampling Rate: We further evaluate `LOFI` against an attacker using an SDR device with

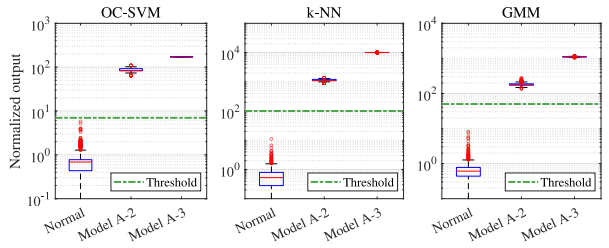


Fig. 8. Output distribution of Vehicle A-1, assuming that the BCM of Vehicle A-2 and Vehicle A-3 are attack devices.

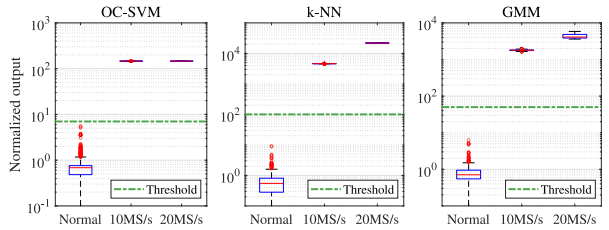


Fig. 9. Output distribution of Vehicle A-1 during a digital attack with high-sample rate.

TABLE V
FEATURE IMPORTANCE AS A FUNCTION OF ATTACK SCENARIOS

Experiment	Rank 1	2	3	4	5	6
§ VI-C	Spec.Centroid$_d$	Spec.Entropy$_d$	Spec.Flatness$_d$	*CFO*	Power$_d$	*SC*
§ VI-D1 § VI-D3	*CFO*	Power$_d$	*SC*	Spec.Entropy$_d$	Spec.Centroid$_d$	Spec.Flatness$_d$
§ VI-D2	*SC*	*CFO*	Power$_d$	Spec.Centroid$_d$	Spec.Flatness$_d$	Spec.Entropy$_d$

a high-sampling rate. We generate attack signals at two sampling rates of 10 MS/s and 20 MS/s, which are supported by the USRP X310 Gigabit Ethernet interface. Five-hundred attack signals are generated per sampling rate. Fig. 9 shows the output distribution of each classifier as a function of a sampling rate when Vehicle A-1 is used as the target vehicle. For a sampling rate of 10 MS/s, the output distribution has a similar pattern as a sampling rate of 2 MS/s. For the sampling rate of 20 MS/s, we see that the output distribution deviates significantly from a legitimate signal. This result leads us to conclude that a SDR device with a high-sampling rate is not able to generate an LF-band signal whose features are similar to a valid signal. Note that LOFI has an average FPR of 0 % and an average FNR of 0 % for all algorithms.

E. Feature Importance

We further analyzed the importance of individual features according to the attack scenarios. Employing the Relieff algorithm [64], we ranked the features as a function of the attack detection experiments as shown in Table V. The Relief algorithm evaluates the rank of features based on the distance between neighbors. For the analog attacker (Section VI-C), Spec.Centroid$_d$, Spec.Entropy$_d$, and Spec.Flatness$_d$ are effective to detect the attack. Based on Table II, we can analyze that the different channel characteristics (i.e., path loss) in ISM band allows LOFI to detect the analog attacker. For the SDR-equipped digital attacker (Section VI-D1), LOFI can effectively differentiate the attack signals using the *CFO* feature. This is because the attacker device has different LO clock offset compared to legitimate BCM. Even when the sample rate increases (Section VI-D3), the rank of importance was preserved. For a digital attack equipped with the same BCM (Section VI-D2), the *CFO* and *SC* have a major role in detecting the attack. This is because of the SC difference between the legitimate BCM and attack device (i.e., BCM of other vehicles).

VII. CONCERNS FOR PRACTICALITY

We additionally evaluate LOFI when it comes to its practicality. First, we evaluate whether LOFI still verifies a legitimate signal under NLoS channel conditions (Section VII-A) or other environmental changes (Section VII-B). Next, we analyze the execution time (Section VII-C) and energy consumption (Section VII-D) required to implement LOFI on a low-cost embedded device. Finally, we evaluate attack detection performance assuming the presence of a strong attacker who can impersonate legitimate LF-band signals.

A. Non-Line-of-Sight Channel Conditions

The PKES system enables a driver to passively unlock car doors without removing the key fob from a storage location, such as a backpack or a pocket. Depending on the location of the key fob, the characteristics of the signal vary due to the multipath channel environment; accordingly, the features used in LOFI can be distorted. If the classifier considers the signals from the legitimate vehicle as belonging to an attack, LOFI raises a false alarm (i.e., does not respond to the vehicle), which hinders the usability of the PKES system. To acquire the signals under the NLoS channel, we measure 500 LF-band signals from a vehicle when the receiver antenna is located inside both a pocket and a backpack. As in previous experiments, features extracted from these signals were used as testing data. Fig. 10 shows the FPR and FNR curves of the classifiers as functions of a threshold Γ. In Fig. 10(a), we observe that an average FPR of 1.44 % and an FNR of 0 % are produced from Vehicle A. Fig. 10(b) illustrates the clear difference between the output distribution of a normal signal and an attack signal, in that the average FPR of 0 % and average FNR of 0 % are associated with Vehicle B. In Fig. 10(c), an average FPR of 1.32 % and average FNR of 0.03 % are produced with Vehicle C.

B. Environmental Variations

The PKES system periodically transmits an LF-band signal to scan whether the corresponding key fob is in the vicinity. Therefore, the PKES system should be available even when the engine is shut off, as the BCM module is connected to a constant power source. The battery condition worsens if the engine is not started and the vehicle sits idle for an extended period of time. Depending on the battery condition, the characteristics of the features used in LOFI may change. To evaluate LOFI for this aging scenario, we supply power at different voltage levels ranging from 12.0 to 10.0 V via a power supplier. Note that classifier training signals are

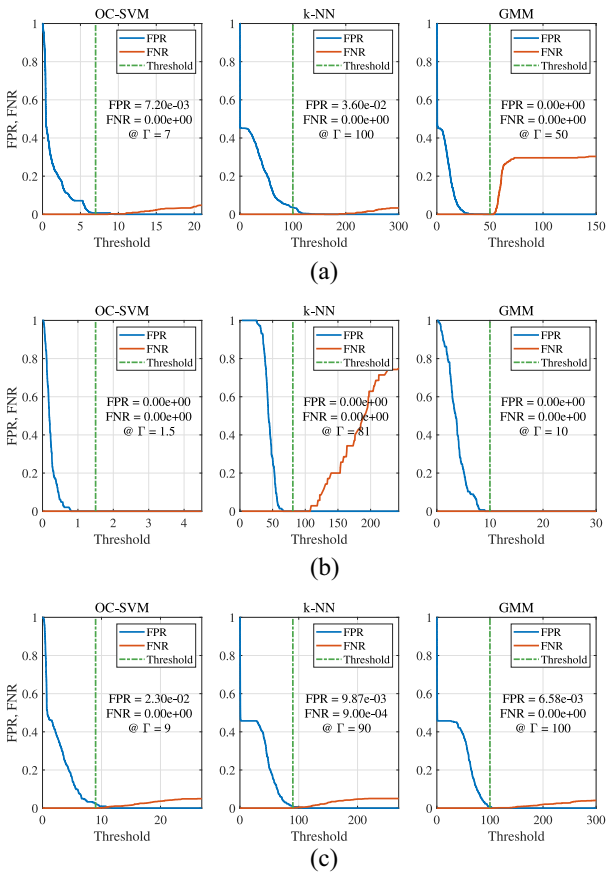


Fig. 10. Attack detection performance under NLoS channel conditions in each algorithm and vehicle model. (a) FPR and FNR results of Vehicle A-1. (b) FPR and FNR results of Vehicle B. (c) FPR and FNR results of Vehicle C.

generated from a genuine battery with a voltage level of 12.4 V. In addition, there was a 9.9 °C temperature difference when we collect training data (i.e., 15.7 °C) and testing data (i.e., 5.8 °C) over six days. In our evaluation, we measure 1,000 LF-band signals per voltage level.

Fig. 11 shows the output distribution of each classifier as a function of voltage level. In all classifiers, we observe that more outliers are given in comparison to the 12.4-V level results. However, since we set a relaxed value for a threshold before (in Table IV), `LOFI` can effectively embrace feature variations caused by the battery level and temperature variations. In our results, `LOFI` shows an FPR of 0 % across three algorithms. Even though battery voltage and temperature differences carry a negligible impact on our experimental setup, there may be incremental feature variations due to environmental changes, aging of electronic components, and dynamic channel conditions (e.g., a busy street). Therefore, it is required to properly manage the records of the feature set and classifier parameters to cope with potential performance degradation. We discuss the details of how `LOFI` can cope with feature variation in Section VIII.

C. Execution Time

We implement feature extraction and classification algorithms on the Arduino Nano board that is a low-performance device. The board is integrated with the ATmega328P processor, which is an 8-bit microcontroller of a 16 MHz clock cycle. Table VI shows the execution time for feature extraction and verification on the Arduino Nano. Through our evaluation, we repeat each execution 100 times and calculate average time. Regarding *SC* and *CFO*, it can be seen that the shortest time is needed for them because they are computed by simply searching a maximum value. For *Spectral.Flatness*, the longest execution time is needed because the *n*th root computation on each sample is required. In addition, time it takes for feature extraction depends on the length of a preamble field of a LF-band signal from a vehicle. Accordingly, it can be seen that the feature extraction time is different according to a vehicle model. For the implementation of OC-SVM, we first use the Python code provided by *scikit-learn* and then we convert the Python code to C-language code by employing the *micromlgen* repository [65]. For GMM and *k*-NN algorithm, we implement C-language codes. As the number of support vectors depends on a training data set, it can be seen that the time for OC-SVM is different according to vehicle models. In our evaluation, the number of 20 to 25 support vectors were observed during the training phase.

From this result, it can be seen that the key fob can verify newly received signals in less than 100 ms. However, the processing delay of 100 ms may make the user feel a lag, which hinders the convenience of the PKES system. To minimize driver-experienced processing delays, `LOFI` starts to verify the LF-band signal as soon as the driver approaches the vehicle, and it completes the process before the driver pulls open a door handle or presses a button on the door. Since it takes a few seconds for the driver to approach the vehicle and pull open a door handle, `LOFI` can efficiently verify the LF-band signal without posing an inconvenience to the driver.

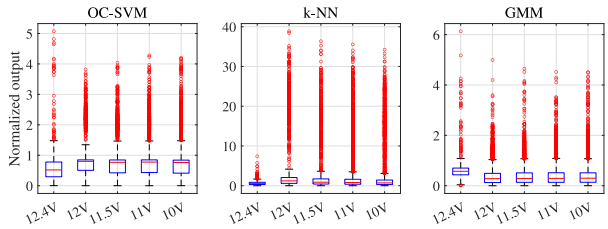


Fig. 11. Output distribution of classifiers.

TABLE VI
EXECUTION TIME

Process	Execution Time		
	Vehicle A	Vehicle B	Vehicle C
SC	4 µs	3 µs	3 µs
CFO	4 µs	3 µs	3 µs
Power_d	4.38 ms	2.92 ms	3.4 ms
$\text{Spectral.Entropy}_d$	145 µs	103 µs	117 µs
$\text{Spectral.Centroid}_d$	296 µs	208 µs	237 µs
$\text{Spectral.Flatness}_d$	55.2 ms	38.53 ms	45.5 ms
Verification (OC-SVM)	5.44 ms	5.69 ms	5.12 ms
Verification (k-NN)	16.14 ms	16.69 ms	15.98 ms
Verification (GMM)	282 µs	280 µs	286 µs

D. Energy Consumption

To understand how much energy `LOFI` requires, we employ a power monitoring device [66], by which we measure the

power difference between the idle state and the verification phase of LOFI implemented in the Arduino Nano. In our results, 5.07 mJ is additionally consumed for the verification phase. We also measure the energy consumed by the key fob of the Vehicle A-1. During normal operation, the key fob receives LF-band signals from the vehicle and transmits UHF-band signals back. The key fob for this normal operation consumes an average of 30 mJ in energy. During modulation and amplification, when UHF-band signals are transmitted, the average energy consumption of the key fob is measured much higher than that of the Arduino Nano. This result shows that LOFI can induce an additional 16.58 % in energy consumption. When LOFI is implemented on the key fob's MCU, however, we expect that the energy consumption is more highly optimized.

E. Against Feature Impersonation Attacker

In many real car theft cases, thieves have carried out relay attacks targeting parked vehicles in front of detached houses at night [5], [6], [7], [8]. Since many people store their key fobs near the front door of their homes, attackers are able to inject malicious LF-band signals into the key fob and trigger a response to the corresponding vehicle. Therefore, the location of the key fob can affect the attack success rate under real car theft scenarios. We evaluate the attack detection performance of LOFI under practical attack scenarios while assuming *the role of the strongest imagined attacker who could perfectly impersonate a legitimate signal*. In this specific attack scenario, we place the key fob in various locations inside of a residence while the attacker injects malicious signals from outside.

1) Setup: We employ a physics-based ray-tracing simulation module [67] to obtain power delay profiles that represent signal strength as a function of a time delay under indoor channel conditions. We model the attacker's antenna locating in front of the door at a height of 1.5 m. We set the relative permittivity of each wall to 6, which is typically used for dry concrete. For simulation parameters, we assume that the attacker could employ a high-power amplifier that provides a transmit power of 30 W [68]. In addition, we set the antenna gain of the transmitter and receiver to −50 dB, which was used in the experiment for a near field channel model [69], [70]. Next, we modify the path loss formula to a near-field path loss model [69] and computed a power delay profile at 1600 measurement points inside the house. Based on this simulation setup, we simulate the attack signals at each point.

For the evaluation, we set the target classifier that is trained with legitimate signals from Vehicle A-1. The attack signals were simulated through convolution operation between a legitimate signal and power delay profile of each location computed from the ray-tracing model. Without applying LOFI, we consider the attack success case (i.e., false negative) if an attack signal is decoded properly. For the attack detection, we perform the feature extraction on each attack signal and verifies it based on threshold in previous experiments.

2) Results: Fig. 12 shows the bivariate histogram as a function of an FNR and a distance from a front door. When LOFI is not applied, we consider a false negative (attack success) case as the case that a bit sequence is properly decoded—key fob verifies the attack signal and transmits a valid response message. As it may seem intuitive, we observe that as the distance from the front door increases, FNRs decrease because the key fob failed to decode the attack signal due to indoor channel conditions, such as signal attenuation and multipath components. Without applying LOFI, FNRs remain above 95 % within a 2-m distance. For longer distances, FNRs begin to degrade and drop to less than 5 %, beginning at 2.3 m. Based on the distribution of the histogram, we compute an expected attack success distance as a sum of the product of an FNR and the distance where the corresponding FNR is obtained (i.e., $\mathbb{E}[d] = \sum_{i=1}^{1600} \mathrm{FNR}_i \cdot d_i$, where FNR_i and d_i refer to FNR and distance of each location).

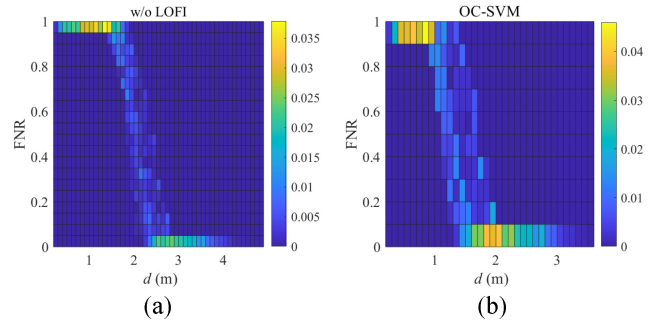


Fig. 12. Bivariate histogram as a function of FNR and the distance from a front door: (a) Before applying LOFI and (b) after applying LOFI using OC-SVM.

In result, the expected attack success distance without LOFI is given by 1.27 m. After LOFI is applied, the attack success rate starts to degrade at 1.4 m in all three algorithms and drops to less than 10 % at distances more than 1.6 m. When LOFI is applied, regardless of classification algorithms, an expected attack success distance of 0.87 m is given. These results indicate a 32 % decrease compared when LOFI is not applied. Even if the attacker perfectly impersonates the signal characteristics, the path loss is an inevitable impairment factor in wireless communication. This result indicates that a feature impersonation attacker can only successfully carry out an attack when located within 1.27 m of the key fob. Previously, it was widely accepted knowledge that analog attackers using cable could successfully inject malicious LF-band signals to the key fob at the distance of 8 m [1].

VIII. Discussion and Future Work

A. Potential Attacks Against PKES System

Although our attack model encompasses all possible attacks against PKES systems that have been introduced so far, potential attacks would be possible. In general, RF signals can be generated employing either analog RF components or digital signal processing techniques. For this reason, we considered analog and digital attackers in Section IV-A and conducted evaluations according to the attacker model. Even under a potential attack scenario, the attacker should generate attack signals employing either analog RF components or a digital signal processor. Therefore, the selected features for attack detection will still effectively detect potential attacks.

Evaluation results on the feature importance (Section VI-E) provide concrete reasoning as to why each feature is affected during the attacks. Therefore, it can be expected that LOFI can effectively detect potential attacks against PKES system.

B. Generality and Robustness

In general, there are infinitely large numbers of NLoS channel conditions in wireless communication systems. As a result, there are numerous studies on channel model for various wireless communication systems, which relies on the stochastic channel model [71], [72], [73], [74]. However, a public channel model for PKES systems is currently unavailable. Fortunately, compared to the general wireless communication system, in the PKES system, the variation of the channel is bounded since the driver stands in front of the vehicle door when he/she locks/unlocks the vehicle. As a result, we have assumed the NLoS channel in the PKES system, focusing on the application scenarios to show the generality of our training data set. Therefore, we believe that building a wireless channel model for PKES system will provide generalized evaluation criteria for both research community and industry companies.

In addition, although we consider potential feature variations due to temperature fluctuations and battery aging—major environmental factors that impact the operation of RF devices, we recognize additional factors degrading the performance of LOFI, which are not covered in this article. To cope with potential feature variations, retraining a machine learning model can be one of the suitable approaches. For Tesla Model X, the BCM manages the key fob to update its firmware via BLE communication [2]. As a response to the OTA attack demonstrated in Section III, Tesla simply updated the vulnerable firmware of the key fob while driving without requiring user interaction [2]. For LOFI, the key fob can convey an extracted feature set to the (BCM) of the vehicle after verifying a received LF-band signal employing ISM-band communication (Fig. 3 in Section IV). When there is a certain amount of feature variation (or verification count), the BCM trains the new model and conveys it to the key fob. In terms of computational cost for training a new model, the OC-SVM model uses the sequential minimal optimization (SMO) algorithm, which has a time complexity of $\mathcal{O}(nd^2)$. In addition, the GMM is trained using the EM algorithm, which has a time complexity of $\mathcal{O}(nd^3)$, where n refers to a number of data and d refers to a number of features in both cases [75], [76]. For the k-NN model that searches for the nearest neighbors during prediction, it is only necessary to update the feature set; that is, the model does not need to be trained (i.e., time complexity of $\mathcal{O}(1)$).

IX. Conclusion

In this article, we presented LOFI, a subauthentication system that supports manufacturer-installed PKES systems, which was designed to analyze low-level characteristics from physical-layer signals for the detection of various attacks. We conducted a series of experiments to evaluate LOFI, for which we considered environmental concerns, including NLoS conditions and battery aging. As a result, we showed that LOFI properly operates even under changing environments. We analyzed which features mainly contribute to detection of an attack per attack type, and moreover, based on the correlation between RF impairment factors and the features used in LOFI, we provided formal reasoning as to why each feature was affected during an attack. Finally, through a ray-tracing simulation, we evaluated the maximum distance at which an attacker, who perfectly impersonates the features used in LOFI. Furthermore, through implementation on a low-cost embedded board, we showed that LOFI enhances the security of the PKES system in practice.

References

[1] A. Francillon, B. Danev, and S. Capkun, "Relay attacks on passive keyless entry and start systems in modern cars," in *Proc. Netw. Distrib. Syst. Secur. Symp. (NDSS)*, 2011, pp. 1–16.

[2] L. Wouters, B. Gierlichs, and B. Preneel, "My other car is your car: Compromising the tesla model X keyless entry system," *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, vol. 2021, no. 4, pp. 149–172, 2021.

[3] L. Wouters, E. Marin, T. Ashur, B. Gierlichs, and B. Preneel, "Fast, furious and insecure: Passive keyless entry and start systems in modern supercars," *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, vol. 2019, no. 3, pp. 66–85, 2019.

[4] L. Wouters, J. Van den Herrewegen, F. D. Garcia, D. Oswald, B. Gierlichs, and B. Preneel, "Dismantling DST80-based immobiliser systems," *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, vol. 2020, no. 2, pp. 99–127, 2020.

[5] "Keyless car thieves caught on cctv stealing £80k mercedes" BBC News. 2019. Accessed: Apr. 28, 2021. [Online]. Available: https://www.bbc.com/news/av/uk-england-leicestershire-47897922

[6] M. Salisbury. "Organised crime gangs target keyless range rovers." FleetPoint. 2021. Accessed: Apr. 28, 2021. [Online]. Available: https://www.fleetpoint.org/telematics-2/tracker/organised-crime-gangs-target-keyless-range-rovers/

[7] "Milton Keynes: CCTV shows 'one-minute keyless car theft.'" BBC News. 2019. Accessed: Apr. 28, 2021. [Online]. Available: https://www.bbc.com/news/av/uk-england-beds-bucks-herts-47203953

[8] "How to stop your car being stolen in 'relay crime theft.'" BBC News. 2017. Accessed: Apr. 28, 2021. [Online]. Available: https://www.bbc.com/news/av/uk-england-birmingham-42142023

[9] G. P. Hancke and M. G. Kuhn, "An RFID distance bounding protocol," in *Proc. 1st Int. Conf. Secur. Privacy Emerg. Areas Commun. Netw.*, 2005, pp. 67–73.

[10] S. Brands and D. Chaum, "Distance-bounding protocols," in *Proc. Workshop Theory Appl. Cryptogr. Techn.*, 1994, pp. 344–359.

[11] B. Danev and S. Capkun, "Transient-based identification of wireless sensor nodes," in *Proc. Int. Conf. Inf. Process. Sens. Netw.*, 2009, pp. 25–36.

[12] A. I. Alrabady and S. M. Mahmud, "Some attacks against vehicles' passive entry security systems and their solutions," *IEEE Trans. Veh. Technol.*, vol. 52, no. 2, pp. 431–439, Mar. 2003.

[13] A. I. Alrabady and S. M. Mahmud, "Analysis of attacks against the security of keyless-entry systems for vehicles and suggestions for improved designs," *IEEE Trans. Veh. Technol.*, vol. 54, no. 1, pp. 41–50, Jan. 2005.

[14] Y. Z. Y. Li. "Chasing cars: Keyless entry system attacks HITBSecConf2017—Amsterdam." Accessed: Dec. 14, 2022. [Online]. Available: https://conference.hitb.org/hitbsecconf2017ams/sessions/chasing-cars-keyless-entry-system-attacks/

[15] K. Ellis and N. Serinken, "Characteristics of radio transmitter fingerprints," *Radio Sci.*, vol. 36, no. 4, pp. 585–597, 2001.

[16] J. Hall, M. Barbeau, and E. Kranakis, "Enhancing intrusion detection in wireless networks using radio frequency fingerprinting," *Commun., Internet, Inf. Technol.*, vol. 1, pp. 1–6, Nov. 2004.

[17] J. Huang, W. Albazrqaoe, and G. Xing, "BlueID: A practical system for Bluetooth device identification," in *Proc. IEEE Conf. Comput. Commun.*, 2014, pp. 2849–2857.

[18] J. Wu, Y. Nan, V. Kumar, M. Payer, and D. Xu, "BlueShield: Detecting spoofing attacks in Bluetooth low energy networks," in *Proc. 23rd Int. Symp. Res. Attacks, Intrus. Defenses ({RAID})*, 2020, pp. 397–411.

[19] D. Zanetti, B. Danev, and S. Capkun, "Physical-layer identification of UHF RFID tags," in *Proc. 16th Annu. Int. Conf. Mobile Comput. Netw.*, 2010, pp. 353–364.

[20] B. Danev, T. S. Heydt-Benjamin, and S. Capkun, "Physical-layer identification of RFID devices," in *Proc. USENIX Secur. Symp.*, 2009, pp. 199–214.

[21] Z. Zhuang et al., "FBSleuth: Fake base station forensics via radio frequency fingerprinting," in *Proc. Asia Conf. Comput. Commun. Secur.*, 2018, pp. 261–272.

[22] K. Youssef, L. Bouchard, K. Haigh, J. Silovsky, B. Thapa, and C. Vander Valk, "Machine learning approach to RF transmitter identification," *IEEE J. Radio Freq. Identif.*, vol. 2, no. 4, pp. 197–205, Dec. 2018.

[23] B. He and F. Wang, "Cooperative specific emitter identification via multiple distorted receivers," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3791–3806, 2020.

[24] G. Shen, J. Zhang, A. Marshall, and J. R. Cavallaro, "Towards scalable and channel-robust radio frequency fingerprint identification for LoRa," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 774–787, 2022.

[25] Y. Qian, J. Qi, X. Kuai, G. Han, H. Sun, and S. Hong, "Specific emitter identification based on multi-level sparse representation in automatic identification system," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 2872–2884, 2021.

[26] B. Danev, D. Zanetti, and S. Capkun, "On physical-layer identification of wireless devices," *ACM Comput. Surv.*, vol. 45, no. 1, pp. 1–29, 2012.

[27] Q. Xu, R. Zheng, W. Saad, and Z. Han, "Device fingerprinting in wireless networks: Challenges and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 94–104, 1st Quart., 2015.

[28] B. Razavi, *Fundamentals of Microelectronics*. Hoboken, NJ, USA: Wiley, 2021.

[29] C.-P. Liang, J.-H. Jong, W. E. Stark, and J. R. East, "Nonlinear amplifier effects in communications systems," *IEEE Trans. Microw. Theory Tech.*, vol. 47, no. 8, pp. 1461–1466, Aug. 1999.

[30] S. P. Yadav and S. C. Bera, "Nonlinearity effect of high power amplifiers in communication systems," in *Proc. Int. Conf. Adv. Commun. Comput. Technol. (ICACACT)*, 2014, pp. 1–6.

[31] Y. Z. Y. Li, "Chasing cars: Keyless entry system attacks," 2017, Accessed: Apr. 16, 2021. [Online]. Available: https://archive.conference.hitb.org/hitbsecconf2017ams/sessions/chasing-cars-keyless-entry-system-attacks/

[32] J. Yu, A. Hu, G. Li, and L. Peng, "A robust RF fingerprinting approach using multisampling convolutional neural network," *IEEE Internet Things J.*, vol. 6, no. 4, pp. 6786–6799, Aug. 2019.

[33] X. Fu et al., "Semi-supervised specific emitter identification method using metric-adversarial training," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10778–10789, Jun. 2023.

[34] G. Oligeri, S. Sciancalepore, S. Raponi, and R. Di Pietro, "PAST-AI: Physical-layer authentication of satellite transmitters via deep learning," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 274–289, 2022.

[35] R. Xie et al., "A generalizable model-and-data driven approach for open-set RFF authentication," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 4435–4450, 2021.

[36] R.-F. Liao et al., "Multiuser physical layer authentication in Internet of Things with data augmentation," *IEEE Internet Things J.*, vol. 7, no. 3, pp. 2077–2088, Mar. 2020.

[37] T. Tian et al., "Transfer learning-based radio frequency fingerprint identification using ConvMixer network," in *Proc. IEEE Glob. Commun. Conf.*, 2022, pp. 4722–4727.

[38] Q. Yang and L. Huang, *Inside Radio: An Attack and Defense Guide*. Singapore: Springer, 2018.

[39] K. Joo, W. Choi, and D. H. Lee, "Hold the door! Fingerprinting your car key to prevent keyless entry car theft," in *Proc. 27th Annu. Netw. Distrib. Syst. Secur. Symp.*, 2020, pp. 1–18.

[40] J. Choi, H.-Y. Yang, and D.-H. Cho, "TEMPEST comeback: A realistic audio eavesdropping threat on mixed-signal SoCs," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2020, pp. 1085–1101.

[41] Y. Cheng, X. Ji, J. Zhang, W. Xu, and Y.-C. Chen, "Demicpu: Device fingerprinting with magnetic signals radiated by CPU," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2019, pp. 1149–1170.

[42] W. Choi, K. Joo, H. J. Jo, M. C. Park, and D. H. Lee, "VoltageIDS: Low-level communication characteristics for automotive intrusion detection system," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 8, pp. 2114–2129, 2018.

[43] J. Zhang, Z. Wang, X. Ji, W. Xu, G. Qu, and M. Zhao, "Who is charging my phone? Identifying wireless chargers via fingerprinting," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 2992–2999, Feb. 2021.

[44] M. Christ, N. Braun, J. Neuffer, and A. W. Kempa-Liehr, "Time series feature extraction on basis of scalable hypothesis tests (tsfresh—A python package)," *Neurocomputing*, vol. 307, pp. 72–77, Sep. 2018.

[45] M. F. Triola, W. M. Goodman, R. Law, and G. Labute, *Elementary Statistics*, vol. 10. Boston, MA, USA: Pearson, 2014.

[46] M. E. Ahmed, I.-Y. Kwak, J. H. Huh, I. Kim, T. Oh, and H. Kim, "Void: A fast and light voice liveness detection system," in *Proc. 29th USENIX Security Symp. (USENIX Security 20)*, 2020, pp. 2685–2702.

[47] G. Zhang, X. Ji, X. Li, G. Qu, and W. Xu, "EarArray: Defending against DolphinAttack via acoustic attenuation," in *Proc. NDSS*, 2021, pp. 1–14.

[48] C. Wu, J. Chen, K. He, Z. Zhao, R. Du, and C. Zhang, "EchoHand: High accuracy and presentation attack resistant hand authentication on commodity mobile devices," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2022, pp. 2931–2945.

[49] A. H. Celdrán, P. M. S. Sánchez, C. Feng, G. Bovet, G. M. Pérez, and B. Stiller, "Privacy-preserving and Syscall-based intrusion detection system for IoT spectrum sensors affected by data falsification attacks," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 8408–8415, May 2023.

[50] T. Kang et al., "Anti-spoofing for fingerprint recognition using electric body pulse response," *IEEE Internet Things J.*, vol. 11, no. 4, pp. 5993–6006, Feb. 2024.

[51] S. S. Khan and M. G. Madden, "A survey of recent trends in one class classification," in *Proc. 20th Irish Conf. AICS*, 2010, pp. 188–197.

[52] D. M. J. Tax, "One-class classification: Concept learning in the absence of counter-examples," Dept. Electr. Eng., Math. and Comput. Sci., Technische Univ., Delft, The Netherlands, Rep., 2002.

[53] S. S. Khan and M. G. Madden, "One-class classification: Taxonomy of study and review of techniques," *Knowl. Eng. Rev.*, vol. 29, no. 3, pp. 345–374, 2014.

[54] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv. (CSUR)*, vol. 41, no. 3, pp. 1–58, 2009.

[55] C. A. Balanis, *Antenna Theory: Analysis and Design*. Hoboken, NJ, USA: Wiley, 2015.

[56] "Automobile PKE low frequency receiver inductor keyless entry inductance coil for car smart key 3d1212 722j." ZSUCORE. 2020. Accessed: Jan. 27,2021. [Online]. Available: https://www.aliexpress.com/item/32840317868.html?spm=a2g0s.9042311.0.0.7a7e4c4dlZmMnx

[57] (Nooelec Inc., Wheatfield, NY, USA). *Nooelec—Ham It Up v1.3 barebones—HF Upconverter*. (2020). Accessed: Jan. 27, 2021. [Online]. Available: https://www.nooelec.com/store/ham-it-up.html

[58] Giorgio. "Feature selection library—File exchange—MATLAB central." 2021. Accessed: Jan. 5, 2021. [Online]. Available: https://www.mathworks.com/matlabcentral/fileexchange/56937-feature-selection-library

[59] (Minicircuit, Brooklyn, NY, USA). *ZX05-U432H-S+*. (2021). Accessed: Apr. 21, 2021. [Online]. Available: https://www.minicircuits.com/pdfs/ZX05-U432H-S+.pdf

[60] (Minicircuit, Brooklyn, NY, USA). *ZX05-43-S+*. (2021). Accessed: Apr. 21, 2021. [Online]. Available: https://www.minicircuits.com/pdfs/ZX05-43-S+.pdf

[61] "Adf4355-3 datasheet and product info analog devices," Analog Devices, Inc., Wilmington, MA, USA, 2021. Accessed: Apr. 21, 2021. [Online]. Available: https://www.analog.com/en/products/adf4355-3.html

[62] "LFTX daughterboard 0-30 MHZ TX | Ettus research, a national instruments brand | the leader in software defined radio (SDR)." EttusResearch. 2020. Accessed: Feb. 13, 2021. [Online]. Available: https://www.ettus.com/all-products/lftx/

[63] B. Danev, H. Luecken, S. Capkun, and K. El Defrawy, "Attacks on physical-layer identification," in *Proc. 3rd ACM Conf. Wireless Netw. Secur.*, 2010, pp. 89–98.

[64] K. Kira and L. A. Rendell, "The feature selection problem: Traditional methods and a new algorithm," in *Proc. 10th Nat. Conf. Artif. Intell.*, 1992, pp. 129–134.

[65] "Eloquentarduino/micromlgen: Generate C code for microcontrollers from python's sklearn classifiers." eloquentarduino. 2021. Accessed: Apr. 28, 2021. [Online]. Available: https://github.com/eloquentarduino/micromlgen

[66] "Purchase high voltage power monitor | Monsoon Solutions, inc. | Bellevue." Accessed: Sep. 27, 2021. [Online]. Available: https://www.msoon.com/online-store

[67] S. Hosseinzadeh (MathWorks, Inc., Natick, MA, USA). *3D Ray Tracing for Indoor Radio Propagation*. (2021). Accessed: Sep. 9, 2021. [Online]. Available: https://mathworks.com/matlabcentral/fileexchange/64695-3d-ray-tracing-for-indoor-radio-propagation

[68] (Mini-Circuits Corp., Brooklyn, NY, USA). *Lzy-22+ High Power Amplifier, 0.1–200 MHZ, 50Ohm*. (2021). Accessed: Aug. 28, 2021. [Online]. Available: https://www.minicircuits.com/pdfs/LZY-22+.pdf

[69] (IEEE, New York, NY, USA). *Near Field Channel Model*. (Oct. 2004). Accessed: Mar. 30, 2021. [Online]. Available: https://www.ieee802.org/15/pub/2004/15-04-0417-02-004a-near-field-channel-model.pdf

[70] H. G. Schantz, "Near field propagation law & a novel fundamental limit to antenna gain versus size," in *Proc. IEEE Antennas Propag. Soc. Int. Symp.*, 2005, pp. 237–240.

[71] A. F. Molisch et al., *IEEE 802.15. 4a Channel Model-Final Report*, IEEE, New York, NY, USA, document 802.1504-0062-02-004a, 2004.

[72] A. A. Khuwaja, Y. Chen, N. Zhao, M.-S. Alouini, and P. Dobbins, "A survey of channel modeling for UAV communications," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2804–2821, 4th Quart., 2018.

[73] D. S. Baum, J. Hansen, J. Salo, G. Del Galdo, M. Milojevic, and P. Kyösti, "An interim channel model for beyond-3G systems: Extending the 3GPP spatial channel model (SCM)," in *Proc. IEEE 61st Veh. Technol. Conf.*, 2005, pp. 3132–3136.

[74] J.-P. Kermoal, L. Schumacher, K. I. Pedersen, P. E. Mogensen, and F. Frederiksen, "A stochastic MIMO radio channel model with experimental validation," *IEEE J. Sel. Areas Commun.*, vol. 20, no. 6, pp. 1211–1226, Aug. 2002.

[75] J. Platt, "Sequential minimal optimization: A fast algorithm for training support vector machines," Microsoft, Redmond, WA, USA, Rep. MSR-TR-98-14, 1998.

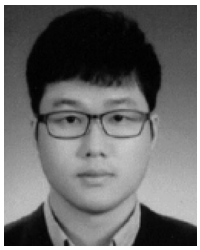
[76] R. C. Pinto and P. M. Engel, "A fast incremental Gaussian mixture model," *PLoS ONE*, vol. 10, no. 10, 2015, Art. no. e0139931.

Kyungho Joo received the B.S. degree from the College of Information and Communication, Korea University, Seoul, South Korea, in 2016, and the M.S. and Ph.D. degrees in information security from Korea University in 2018 and 2024, respectively.

From 2024, he has been an Assistant Professor with Soongsil University, Seoul, Republic of Korea. His research interests include wireless security and automotive security.

Hyo Jin Jo received the B.S. degree in industrial engineering and the Ph.D. degree in information security from Korea University, Seoul, South Korea, in 2009 and 2016, respectively.

He was a Postdoctoral Researcher with the Department of Computer and Information Systems, University of Pennsylvania, Philadelphia, PA, USA, from 2016 to 2018. He is currently an Associate Professor with the Graduate School of Information, Yonsei University, Seoul. His research interests include applied cryptography, security and privacy for ad-hoc networks, and IoT/CPS security.



Wonsuk Choi received the B.S. degree in mathematics from the University of Seoul, Seoul, South Korea, in 2008, and the M.S. and Ph.D. degrees in information security from Korea University, Seoul, in 2013 and 2018, respectively.

He was a Postdoctoral Researcher with the Institute of Cyber Security and Privacy, Korea University. In 2023, he was an Assistant Professor with the Graduate School of Information Security, Korea University. His research interests include security for body area networks, usable security, applied cryptography, and smart car security.